**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 113 · Comparing Algorithms and Consolidation

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. How does merge sort combine two sorted lists?

Q2. Which storage technology does a DVD use?
A. Optical
B. Magnetic
C. Solid-state
D. Volatile RAM

Q3. “FROM is the clause that specifies the required output columns.” Correct this. Explain why.

Q4. Code has a missing bracket. “A syntax error is a wrong result from code that follows the language's rules.” Correct this. What needs fixing?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Searching — Locating a target in a data set.

Sorting — Arranging items into a specified order.

Prerequisite — Condition required for a method to work correctly.

Divide and conquer — Solving a problem by splitting into smaller parts.

Comparison — Checking relative values to decide the next operation.

## New learning

Search versus sort: Searching locates a target; sorting rearranges values into a specified order.

Choose a search: Linear works on unsorted data; binary needs sorted data but often makes fewer comparisons for large lists.

Compare sorting approaches: Bubble swaps adjacent out-of-order pairs; insertion places the next key into a sorted prefix; merge divides and combines ordered sublists.

Explain from evidence: For examination questions, name the method, trace the data and justify any prerequisite or limitation.

## Worked example

A school stores an unsorted list of pupil IDs. Choose a suitable search, then explain what would change if the list were sorted. [5 marks]

1. Identify the starting condition

2. The original list is unsorted.

3. Linear search works without sorting and checks entries in sequence.

4. If sorted, binary search could discard half of the active range after each comparison and generally need fewer checks for large lists. [5]

# We Do · Practise with guidance

### Guided 1

A sorted array contains 128 items. Which search is appropriate and why? [4]

Worked method: Binary search [1] as sorting prerequisite is met [1]; midpoint comparisons halve the range [1], generally using fewer comparisons than linear search for large data [1].

### Guided 2

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

A student says bubble and insertion sort are identical. Compare the two. [4]

Worked method: Bubble compares/swaps neighbouring elements [1] over repeated passes [1]; insertion builds a sorted prefix [1] by shifting and inserting a key [1].

**Guided 3**

Explain the difference between searching for a value and sorting data. [3]

Worked method: Search aims to locate a value/index [1]; sort reorders all items by a key [1]; binary search may require sorting first [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Predict which search algorithm may be used for a sorted list of common network protocols.

```
array protocols = ["DNS","HTTP","HTTPS","SMTP"]
sorted = true
if sorted then
    print("Binary search possible")
else
    print("Use linear search")
endif
```

Predict the output.

Change sorted to false.

Explain the functions of DNS and HTTPS, and why binary search needs ordered data.

### Question 1

Choose a search algorithm for an unsorted list and explain the choice. [3] Source: Original OCR J277-style practice question.

### Question 2

State the characteristic action of each sorting algorithm: bubble, insertion and merge. [3] Source: Original OCR J277-style practice question.

### Question 3

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Explain why binary search may outperform linear search on a large sorted array. [4]
Source: Original OCR J277-style practice question.

## Exit Ticket · Answer before checking

Binary search can safely discard half only when data is sorted.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

## Page 1 onwards · Do Now

Q1. Compare the next item in each list. Take the smaller. Repeat, then add any items left over.

Q2. A — Optical

Q3. The table to get data from.

Q4. A syntax error breaks the language's rules and stops the affected code being translated or run. Fix it, then trace and test the logic.

## I Do / We Do pages · Guided practice

Q1. Binary search [1] as sorting prerequisite is met [1]; midpoint comparisons halve the range [1], generally using fewer comparisons than linear search for large data [1].

Q2. Bubble compares/swaps neighbouring elements [1] over repeated passes [1]; insertion builds a sorted prefix [1] by shifting and inserting a key [1].

Q3. Search aims to locate a value/index [1]; sort reorders all items by a key [1]; binary search may require sorting first [1].

## You Do pages · Code First

Expected result: The sorted flag is true, so Binary search possible. If false, Use linear search. DNS resolves domain names to IP addresses; HTTPS secures web traffic using TLS encryption. Binary search depends on sorted input so it can safely eliminate half the remaining range. This code chooses an approach; it does not implement the search.

## You Do pages · Independent questions

Q1. Linear search [1]; compares items sequentially [1] without needing the list sorted [1].

Q2. Bubble: adjacent swaps [1]; insertion: insert key into sorted prefix [1]; merge: split then merge ordered sublists [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Q3. Binary chooses midpoint [1] and discards half each unsuccessful check [1]; linear may inspect elements individually [1], so binary generally uses fewer checks for large sorted arrays [1].

### Exit Ticket · Model answer

True

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**